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Satellite equipment and services

Satellite payloads consist of rubidium atomic clocks and other electronic equipment to generate navigation signals. Employing both L5 (1176.45MHz) and S band (2492.028MHz), the system provides two kinds of services, namely, Standard Positioning Service (SPS) and Precision Service (PS). The SPS signal is modulated by a 1MHz Binary Phase Shift Key (BPSK) modulation, whereas PS uses Binary Offset Carrier (BOC).

GNSS SIMULATOR



SATELLITE DETAILS

Name	Launch date	Launch vehicle
IRNSS-1A	July 1, 2013	PSLV-C22
IRNSS-1B	April 4, 2014	PSLV-C24
IRNSS-1C	October 16, 2014	PSLV-C26
IRNSS-1D	March 28, 2015	PSLV-C27
IRNSS-1E	January 20, 2016	PSLV-C31
IRNSS-1F	March 10, 2016	PSLV-C32
IRNSS-1G	April 28, 2016	PSLV-C33

ground segment. Three satellites are located in GEO and the other four in inclined GSO. Such an arrangement means continuous radio visibility at all hours for Indian control stations.

"We typically have two signal frequencies to overcome the problem caused by disruption in the signal, which is caused by the ionosphere. This increases the size of the module due to the integration of two separate antennae for the two frequencies. Hence, a solution is to use a single antenna, meaning, a single frequency being used for navigation. Using a single antenna causes very low accuracy and can show a position difference by 10 metres to 15 metres.

"Addition of a second signal reduces this to about five metres to eight metres. Hence, a system integrator could decide on designing the system to reduce system complexity with a single antenna system for low-accuracy needs or placing two anten-

nae for high-accuracy needs," says Dr Vyasraj on using multiple-frequency signals for location tracking.

He adds, "IRNSS systems have been working on adding L1 band for navigation. And very soon we may have three signals to further improve location accuracy for navigation systems." The system uses grid based ionosphere-related corrections for single-frequency users (L5/S) to improve location accuracy.

Testing and development

Before completion and launch of satellites, there is the need for testing the design parameters and the functioning modules. Design simulators have helped play a huge role in this stage of the development. "We have a simulator from Accord Software capable of generating and measuring signal conditions upon reception. The device hardware is compatible with American GPS, Russian GLONASS, Indian IRNSS, European Galileo and Chinese BeiDou/BDS.

"The simulator also has support for several SBAS systems. We can modulate and test signal parameters during various stages of transmission. The simulator generates signal distortions similar to the actual conditions and gives results, which can help fine-tune GNSS equipment," says Dr Pratibha B.A., project leader at Accord Software.

For field-testing purposes, there